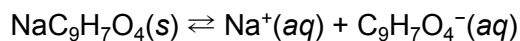
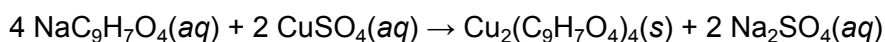


Although acetylsalicylic acid (aspirin) is a well-known and widely used analgesic and anti-inflammatory agent, copper(II) acetylsalicylate, a copper complex formed from aspirin and copper, has also been found to exhibit some distinctive and potentially beneficial pharmacological properties. Formation of copper(II) acetylsalicylate is achieved by dissolving an aspirin salt in water (Reaction 1) and then introducing Cu^{2+} ions which bind to the conjugate base of aspirin, acetylsalicylate ($\text{C}_9\text{H}_7\text{O}_4^-$), as outlined in Reaction 2.



Reaction 1



Reaction 2

Using Reaction 2, a researcher synthesized a sample of copper(II) acetylsalicylate for analysis. A small volume of 0.50 M aqueous copper(II) sulfate (molar mass = 159.6) was mixed into 100 mL of filtered, saturated aqueous sodium acetylsalicylate (molar mass = 202.2) at room temperature and slowly stirred. Formation of copper(II) acetylsalicylate (molar mass = 843.7) was immediately observed as a bright blue precipitate. The precipitate was then isolated by filtration and dried, affording the product complex in high yield.

If the solubility of sodium acetylsalicylate is roughly 120 g per 100 mL of water, what is the approximate molar concentration of the saturated $\text{NaC}_9\text{H}_7\text{O}_4$ solution used in the reaction?

- ☐ A. 0.006 mol/L [6%]
- ☐ B. 0.59 mol/L [20%]
- ✓ ☒ C. 5.9 mol/L [69%]
- ☐ D. 1,200 mol/L [3%]

Correct

69% answered correctly

Explanation:

Saturated solubility	Volume conversion	Molar mass $\text{NaC}_9\text{H}_7\text{O}_4$	Molarity (M)
$\frac{120 \text{ g NaC}_9\text{H}_7\text{O}_4}{100 \text{ mL H}_2\text{O}}$	$\times \frac{1000 \text{ mL}}{1 \text{ L}}$	$\times \frac{1 \text{ mol NaC}_9\text{H}_7\text{O}_4}{202.2 \text{ g NaC}_9\text{H}_7\text{O}_4}$	$= 5.9 \frac{\text{mol}}{\text{L}}$

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A **saturated solution** contains the **maximum amount** of a compound that will dissolve in a given volume of solvent under the given conditions. **Solubility** measurements are often stated as the mass of compound that will dissolve per 100 g of solvent. For water (density = 1 g/mL) as a solvent, this can be expressed as 100 mL. Applying unit conversions, the concentration of a saturated solution can be

expressed in units of **molarity**, which is defined as moles of solute per liter of solution (mol/L).

At saturation, the $\text{NaC}_9\text{H}_7\text{O}_4$ solution contains 120 g of $\text{NaC}_9\text{H}_7\text{O}_4$ in every 100 mL of water. This ratio equates to a concentration of 1.20 g/mL. To convert this into molarity, the volume must be converted from milliliters to liters (1000 mL = 1 L), and the mass (grams) must be converted into moles using the **molar mass** of $\text{NaC}_9\text{H}_7\text{O}_4$ (202.2 g/mol). Multiplying 1.2 g/mL by the volume conversion and dividing the grams by the molar mass yields a saturating molarity of 5.9 mol/L.

(Choice A) A value of 0.006 M results from converting the mass to moles but failing to convert the unit of volume from milliliters to liters. This gives units of mol/mL instead of mol/L.

(Choice B) A value of 0.59 M results from converting the mass to moles but failing to divide by the volume of the solution. This gives units of moles rather than mol/L.

(Choice D) A value of 1200 M results from converting the volume from milliliters to liters but failing to convert the mass to moles. This gives units of g/L, not mol/L.

Educational objective:

A saturated solution contains the maximum amount of compound that will dissolve in a given volume of solvent under the given conditions. Solubility (expressed as grams of solute per 100 g of solvent) can be stated in units of molarity (moles of solute per liter of solvent) using unit conversions and the molar mass of the solute.

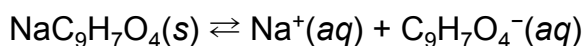
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Subject:
General Chemistry

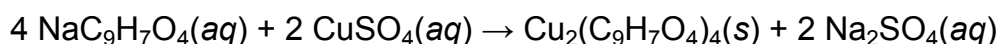
QID:401527

System:
Solutions and Electrochemistry

Although acetylsalicylic acid (aspirin) is a well-known and widely used analgesic and anti-inflammatory agent, copper(II) acetylsalicylate, a copper complex formed from aspirin and copper, has also been found to exhibit some distinctive and potentially beneficial pharmacological properties. Formation of copper(II) acetylsalicylate is achieved by dissolving an aspirin salt in water (Reaction 1) and then introducing Cu^{2+} ions which bind to the conjugate base of aspirin, acetylsalicylate ($\text{C}_9\text{H}_7\text{O}_4^-$), as outlined in Reaction 2.



Reaction 1



Reaction 2

Using Reaction 2, a researcher synthesized a sample of copper(II) acetylsalicylate for analysis. A small volume of 0.50 M aqueous copper(II) sulfate (molar mass = 159.6) was mixed into 100 mL of filtered, saturated aqueous sodium acetylsalicylate (molar mass = 202.2) at room temperature and slowly stirred. Formation of copper(II) acetylsalicylate (molar mass = 843.7) was immediately observed as a bright blue precipitate. The precipitate was then isolated by filtration and dried, affording the product complex in high yield.

What volume of the copper(II) sulfate solution should be used in the reaction to deliver 0.8 g of CuSO_4 into the reaction mixture?

- ☐ A. 1.6 mL [8%]
☐ B. 2.5 mL [13%]
☒ C. 0.01 L [71%]
☐ D. 0.40 L [6%]

Correct

72% answered correctly

Explanation:

Mass needed		Molar mass CuSO_4		CuSO_4 solution molarity		Volume needed
0.8 g CuSO_4	$\times$	$\frac{1 \text{ mol } \text{CuSO}_4}{159.6 \text{ g } \text{CuSO}_4}$	$\times$	$\frac{1 \text{ L solution}}{0.50 \text{ mol } \text{CuSO}_4}$	$=$	0.01 L

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The **molar mass** (molecular weight) is the amount of mass present in 1 mol of a substance, and it relates a given mass to the number of molecules present in a sample. The **molarity** of a solution is the number of moles of a substance present in 1 L of a solution. Therefore, the

molar mass and the molarity relate to each other by the number of moles, and both can be **used as conversion factors** in chemical reaction calculations involving a given mass.

If 0.8 g need to be delivered into the reaction, the number of moles in this amount of mass can be found by dividing by the molar mass of CuSO_4 :

$$(0.8 \text{ g}) / (159.6 \text{ g/mol}) = 0.005 \text{ mol CuSO}_4$$

To find the volume of a 0.50 M solution required to obtain 0.005 mol CuSO_4 for delivery into the reaction, divide the number of moles by the molarity of the solution:

$$(0.005 \text{ mol CuSO}_4) / (0.50 \text{ mol/L}) = 0.01 \text{ L CuSO}_4 \text{ solution}$$

(Choice A) 1.6 mL is the value obtained if the mass is not converted to moles (dividing by molar mass) and is just divided by the molarity. Incomplete cancellation yields units of $\text{g}\cdot\text{L/mol}$ instead of units of volume.

(Choice B) 2.5 mL is the value obtained if the mass is converted to moles but is then multiplied by the molarity (instead of dividing). Incorrect units of mol^2/L result instead of units of volume.

(Choice D) 0.40 L is the value obtained if the mass is not converted to moles and is instead multiplied by the molarity. Units of $\text{g}\cdot\text{mol/L}$ are obtained instead of units of volume.

Educational objective:

The molar mass expresses the mass present in 1 mol of a substance whereas the molarity of a solution expresses the number of moles of a substance dissolved in 1 L of a solution. The molar mass and the molarity relate to each other by the number of moles, and both can be used as conversion factors in chemical reaction calculations.

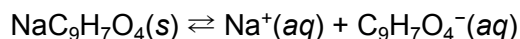
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Subject:
General Chemistry

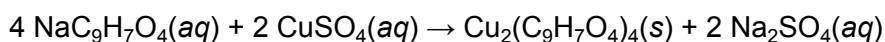
QID:401528

System:
Solutions and Electrochemistry

Although acetylsalicylic acid (aspirin) is a well-known and widely used analgesic and anti-inflammatory agent, copper(II) acetylsalicylate, a copper complex formed from aspirin and copper, has also been found to exhibit some distinctive and potentially beneficial pharmacological properties. Formation of copper(II) acetylsalicylate is achieved by dissolving an aspirin salt in water (Reaction 1) and then introducing Cu^{2+} ions which bind to the conjugate base of aspirin, acetylsalicylate ($\text{C}_9\text{H}_7\text{O}_4^-$), as outlined in Reaction 2.



Reaction 1



Reaction 2

Using Reaction 2, a researcher synthesized a sample of copper(II) acetylsalicylate for analysis. A small volume of 0.50 M aqueous copper(II) sulfate (molar mass = 159.6) was mixed into 100 mL of filtered, saturated aqueous sodium acetylsalicylate (molar mass = 202.2) at room temperature and slowly stirred. Formation of copper(II) acetylsalicylate (molar mass = 843.7) was immediately observed as a bright blue precipitate. The precipitate was then isolated by filtration and dried, affording the product complex in high yield.

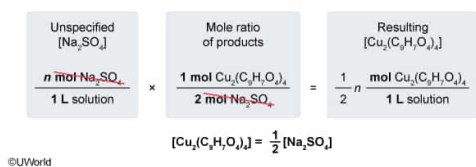
In Reaction 2, just prior to precipitation, what is the ratio of the concentration of copper(II) acetylsalicylate to the concentration of sodium sulfate?

- ☐ A. 1/4 [5%]
☒ B. 1/2 [78%]
☐ C. 2/2 [6%]
☐ D. 2/1 [9%]

Correct

78% answered correctly

Explanation:



Chemical compounds react in definite **proportions** (ratios) during a chemical reaction. The **coefficients** in front of each chemical formula in a balanced chemical reaction indicate the **stoichiometric mole ratios** by which each chemical combines during the reaction. Therefore, mole ratios derived from these coefficients can be used as **conversion factors** in chemical reaction calculations to relate the molar amount of one substance to the molar amount of another in a reaction. Accordingly, these ratios also relate the concentration of one product to that of another for a given reaction volume.

Reaction 2 shows that there will be 2 moles of Na_2SO_4 produced for every 1 mole of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$ produced within the same solution volume. Therefore, the molar concentration of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$ equals half the molar concentration of Na_2SO_4 , and a reaction that produces n moles of Na_2SO_4 per liter of solution will produce $n/2$ moles of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$ per liter of solution. As a result, the $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$

concentration (prior to precipitation) is equal to half the Na_2SO_4 concentration.

(Choice A) A ratio of 1/4 corresponds to the mole ratio between $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$ and the $\text{NaC}_9\text{H}_7\text{O}_4$ reactant instead of between $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$ and Na_2SO_4 .

(Choice C) A ratio of 2/2 corresponds to the mole ratio between the Na_2SO_4 product and the CuSO_4 reactant instead of between $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$ and Na_2SO_4 .

(Choice D) A ratio of 2/1 is the mole ratio of Na_2SO_4 to $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$, rather than the ratio of the mole ratio of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$ to Na_2SO_4 (the ratio has been flipped).

Educational objective:

The coefficients in a balanced chemical reaction indicate the stoichiometric proportions by which each chemical combines during the reaction. Mole ratios derived from these coefficients can be used as conversion factors in chemical reaction calculations to relate the molar amounts of compounds in a reaction.

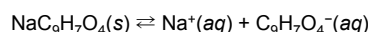
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Subject:
General Chemistry

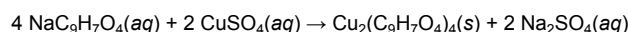
QID:401529

System:
Solutions and Electrochemistry

Although acetylsalicylic acid (aspirin) is a well-known and widely used analgesic and anti-inflammatory agent, copper(II) acetylsalicylate, a copper complex formed from aspirin and copper, has also been found to exhibit some distinctive and potentially beneficial pharmacological properties. Formation of copper(II) acetylsalicylate is achieved by dissolving an aspirin salt in water (Reaction 1) and then introducing Cu^{2+} ions which bind to the conjugate base of aspirin, acetylsalicylate ($\text{C}_9\text{H}_7\text{O}_4^-$), as outlined in Reaction 2.



Reaction 1



Reaction 2

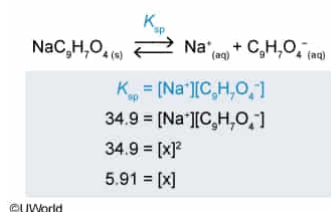
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If the solubility product constant K_{sp} for $\text{NaC}_9\text{H}_7\text{O}_4$ is estimated to be 34.9, what is the approximate acetylsalicylate ion concentration in the saturated $\text{NaC}_9\text{H}_7\text{O}_4$ solution used for the reaction?

- ☐ A. 2.95 M [8%]
☒ B. 5.91 M [57%]
☐ C. 17.4 M [19%]
☐ D. 34.9 M [13%]

Correct
 57% answered correctly

Explanation:



For the dissolution of compound Z into ions X and Y, a general dissolution reaction can be written as:



From the general dissolution reaction above, an equilibrium constant associated with the dissolution process can be written to give the **solubility product constant K_{sp}** :

$$K_{\text{sp}} = [\text{X}]^c [\text{Y}]^d$$

In the K_{sp} expression, $[\text{X}]$ and $[\text{Y}]$ are the molar concentrations of the X and Y ions, and c and d are the respective coefficients of the balanced dissolution reaction. The K_{sp} is the product of the maximum concentrations of each ion possible in the solution under the given conditions. As a result, the K_{sp} correlates to the **limit of solubility** for the compound (a **saturated solution**).

For the dissociation of $\text{NaC}_9\text{H}_7\text{O}_4$, which releases one Na^+ ion and one $\text{C}_9\text{H}_7\text{O}_4^-$ ion, $K_{\text{sp}} = [\text{Na}^+][\text{C}_9\text{H}_7\text{O}_4^-]$. Because 1 mole of Na^+ and 1 mole of $\text{C}_9\text{H}_7\text{O}_4^-$ are released into solution for each mole of $\text{NaC}_9\text{H}_7\text{O}_4$ that dissociates, $[\text{Na}^+]$ and $[\text{C}_9\text{H}_7\text{O}_4^-]$ are always equal.

$$[\text{Na}^+] = [\text{C}_9\text{H}_7\text{O}_4^-] = x$$

As a result, the K_{sp} of $\text{NaC}_9\text{H}_7\text{O}_4$ is:

$$K_{\text{sp}} = [\text{Na}^+][\text{C}_9\text{H}_7\text{O}_4^-] = x^2$$

Therefore, the concentration of the acetylsalicylate ion in a saturated solution is equal to the square root of the K_{sp} :

$$[\text{C}_9\text{H}_7\text{O}_4^-] = \sqrt{K_{\text{sp}}} = \sqrt{34.9}$$

$$[\text{C}_9\text{H}_7\text{O}_4^-] = 5.91 \text{ M}$$

Here, it is unnecessary to perform a rigorous manual calculation of the square root of 34.9. Instead, the correct answer can be identified by approximation. Because $5^2 = 25$, and $6^2 = 36$, the square root of 34.9 must be between 5 and 6.

(Choice A) 2.95 M is half of the molar concentration of the $\text{C}_9\text{H}_7\text{O}_4^-$ ion. In the saturated solution, both the Na^+ ion and the $\text{C}_9\text{H}_7\text{O}_4^-$ ion have a concentration of 5.91 M. The calculation does not require dividing the result in half.

(Choice C) 17.4 M is obtained if the K_{sp} is divided in half rather than taking the square root.

(Choice D) 34.9 is the value of the K_{sp} rather than the concentration of the $C_9H_7O_4^-$ ion.

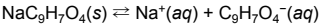
Educational objective:

The solubility product constant K_{sp} represents the limit of solubility for a compound, and it is defined as the product of the molar concentrations of the dissolved species each raised to the power of their respective balanced equation coefficients. This relationship can be used to find molar concentrations of ions in saturated solutions.

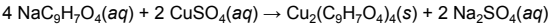
Time spent: 0 sec
Subject:
General Chemistry

QID: 401530
System:
Solutions and Electrochemistry

Although acetylsalicylic acid (aspirin) is a well-known and widely used analgesic and anti-inflammatory agent, copper(II) acetylsalicylate, a copper complex formed from aspirin and copper, has also been found to exhibit some distinctive and potentially beneficial pharmacological properties. Formation of copper(II) acetylsalicylate is achieved by dissolving an aspirin salt in water (Reaction 1) and then introducing Cu^{2+} ions which bind to the conjugate base of aspirin, acetylsalicylate ($\text{C}_9\text{H}_7\text{O}_4^-$), as outlined in Reaction 2.



Reaction 1



Reaction 2

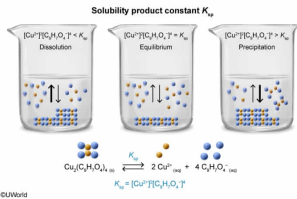
Using Reaction 2, a researcher synthesized a sample of copper(II) acetylsalicylate for analysis. A small volume of 0.50 M aqueous copper(II) sulfate (molar mass = 159.6) was mixed into 100 mL of filtered, saturated aqueous sodium acetylsalicylate (molar mass = 202.2) at room temperature and slowly stirred. Formation of copper(II) acetylsalicylate (molar mass = 843.7) was immediately observed as a bright blue precipitate. The precipitate was then isolated by filtration and dried, affording the product complex in high yield.

The copper(II) acetylsalicylate product immediately precipitates from the reaction solution because:

- ☐ A. $[\text{Cu}^{2+}]^2$ multiplied by $[\text{C}_9\text{H}_7\text{O}_4^-]^4$ is less than the K_{sp} of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$. [22%]
- ☐ B. the sum of $[\text{Cu}^{2+}]^2$ and $[\text{C}_9\text{H}_7\text{O}_4^-]^4$ is less than the K_{sp} of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$. [3%]
- ☒ C. $[\text{Cu}^{2+}]^2$ multiplied by $[\text{C}_9\text{H}_7\text{O}_4^-]^4$ is greater than the K_{sp} of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$. [69%]
- ☐ D. the sum of $[\text{Cu}^{2+}]^2$ and $[\text{C}_9\text{H}_7\text{O}_4^-]^4$ is greater than the K_{sp} of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$. [5%]

Correct
69% answered correctly

Explanation:



The dissolution of compound Z into ions X and Y can be written as:



The solubility product constant K_{sp} for the general dissolution reaction above is given by

$$K_{\text{sp}} = [\text{X}]^c [\text{Y}]^d$$

where $[\text{X}]$ and $[\text{Y}]$ are the molar concentrations of the X and Y ions, and c and d are the respective coefficients of the balanced dissolution reaction. The K_{sp} is the product of the maximum concentrations of each ion possible in the solution under the given conditions. As a result, the K_{sp} represents the **limit of solubility** for the compound (**saturated** solution).

If the concentrations of $[\text{X}]$ and $[\text{Y}]$ are such that $[\text{X}]^c[\text{Y}]^d < K_{\text{sp}}$, the solution has not reached the solubility limit and will permit more ions into solution (dissolve more of Z). Conversely, if the concentrations of $[\text{X}]$ and $[\text{Y}]$ are such that $[\text{X}]^c[\text{Y}]^d > K_{\text{sp}}$, the solubility limit of the solution has been exceeded and the solution will not accept any more ions (no more Z will dissolve). This will cause any ions in **excess of the K_{sp}** to come out of solution as a **precipitate**.

For the dissolution of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$, the maximum solubility of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$ (the K_{sp}) is equal to $[\text{Cu}^{2+}]^2[\text{C}_9\text{H}_7\text{O}_4^-]^4$. Although the numerical value of K_{sp} is not given, the fact that the product immediately precipitates from solution indicates that the concentrations of $[\text{Cu}^{2+}]$ and $[\text{C}_9\text{H}_7\text{O}_4^-]$ result in a value of $[\text{Cu}^{2+}]^2[\text{C}_9\text{H}_7\text{O}_4^-]^4$ that exceeds the K_{sp} of $\text{Cu}_2(\text{C}_9\text{H}_7\text{O}_4)_4$. Therefore, $[\text{Cu}^{2+}]^2[\text{C}_9\text{H}_7\text{O}_4^-]^4 > K_{\text{sp}}$.

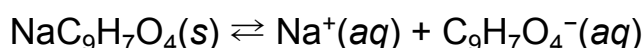
(Choice A) $[\text{Cu}^{2+}]^2[\text{C}_9\text{H}_7\text{O}_4^-]^4 < K_{\text{sp}}$ indicates that the solubility limit of the product has not been exceeded and the product would not precipitate from the solution.

(Choice B and D) Because $K_{\text{sp}} = [\text{Cu}^{2+}]^2[\text{C}_9\text{H}_7\text{O}_4^-]^4$ at saturation, comparisons of K_{sp} with the expression $[\text{Cu}^{2+}]^2 + [\text{C}_9\text{H}_7\text{O}_4^-]^4$ are not meaningful with respect to solubility.

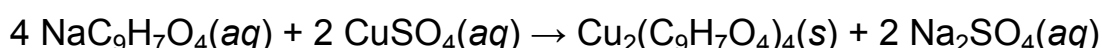
Educational objective:

In a solution formed from the dissolution of a compound into ions, the compound will dissolve until the solution reaches the value of the solubility product constant K_{sp} for the compound. Ion concentrations that exceed the K_{sp} result in precipitation of the excess.

Although acetylsalicylic acid (aspirin) is a well-known and widely used analgesic and anti-inflammatory agent, copper(II) acetylsalicylate, a copper complex formed from aspirin and copper, has also been found to exhibit some distinctive and potentially beneficial pharmacological properties. Formation of copper(II) acetylsalicylate is achieved by dissolving an aspirin salt in water (Reaction 1) and then introducing Cu^{2+} ions which bind to the conjugate base of aspirin, acetylsalicylate ($\text{C}_9\text{H}_7\text{O}_4^-$), as outlined in Reaction 2.



Reaction 1



Reaction 2

Using Reaction 2, a researcher synthesized a sample of copper(II) acetylsalicylate for analysis. A small volume of 0.50 M aqueous copper(II) sulfate (molar mass = 159.6) was mixed into 100 mL of filtered, saturated aqueous sodium acetylsalicylate (molar mass = 202.2) at room temperature and slowly stirred. Formation of copper(II) acetylsalicylate (molar mass = 843.7) was immediately observed as a bright blue precipitate. The precipitate was then isolated by filtration and dried, affording the product complex in high yield.

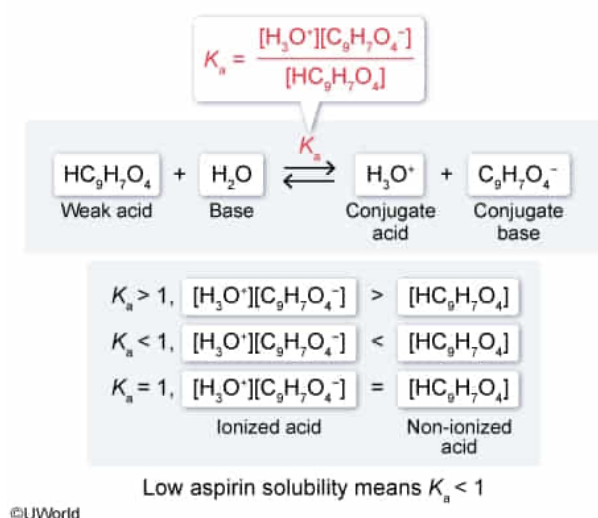
The aqueous solubility of the $\text{NaC}_9\text{H}_7\text{O}_4$ salt used for the reaction is very high ($\approx 1200 \text{ g/L}$), whereas its conjugate acid, $\text{HC}_9\text{H}_7\text{O}_4$ (aspirin), is a weak acid with very low solubility (3 g/L). The acid dissociation constant for aspirin is:

- ☐ A. > 1 . [18%]
- ✓ ☒ B. < 1 . [79%]
- ☐ C. $= 1$. [1%]
- ☐ D. $= 0$.

Correct

79% answered correctly

Explanation:



When an **acid (HA)** is dissolved and ionized in water, a proton (H^+) is released into solution, resulting in the formation of a hydronium ion (H_3O^+) and a **conjugate base** (A^-) of the acid. The equilibrium for this process is quantified by the **acid dissociation constant K_a** . Strong acids have an equilibrium favoring the products (the release of H^+ and A^-) and ionize essentially 100% in aqueous solutions, resulting in a large acid dissociation constant ($K_a > 1$). **Weak acids** have an equilibrium favoring HA, the non-ionized acid, and ionize only to a small extent, resulting in a **small acid dissociation constant ($K_a < 1$)**.

Because aspirin is a weak acid, the K_a value for aspirin is expected to be small ($K_a < 1$). This is consistent with the low solubility reported for aspirin. A strong acid with $K_a > 1$ would ionize to a large extent and would produce highly water-soluble ions, as seen for the ionic sodium salt of aspirin. The low solubility of the neutral aspirin acid indicates a **low extent of ionization** in solution.

(Choice A) $K_a > 1$ indicates high concentrations of ionized acid species (H_3O^+ and A^-) relative to the non-ionized acid HA, and is consistent with a strong acid rather than a weak acid.

(Choice C) $K_a = 1$ would only occur if $[\text{H}_3\text{O}^+][\text{A}^-]$ were equal to the concentration of HA, indicating a greater extent of ionization than is typical for weak acids.

(Choice D) $K_a = 0$ would only occur if no ionization occurs; a compound cannot act as a proton-donating acid if ionization does not occur. Although weak acids ionize only to a small extent, the ionization does occur and is measurably nonzero.

Educational objective:

Strong acids ionize essentially 100% in aqueous solutions, resulting in a large acid dissociation constant ($K_a > 1$), whereas weak acids ionize only to a small extent, resulting in a small acid dissociation constant ($K_a < 1$).

Time spent:0 sec

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Subject:
General Chemistry

System:
Solutions and Electrochemistry